

COMP3052.SEC Computer Security

Session 01: Introduction to COMP3052.SEC



ACKNOWLEDGEMENTS

- Some of the materials we use this semester may come directly from previous teachers of this module, and other sources ...
- Thank you to (amongst others):
 - Michel Valstar, Milena Radenkovic, Mike Pound, Dave Towey,...

OVERVIEW

- Convenor & Teacher Information
- Module Information
- Assessment
- Motivation for the Module
- Module Contents
- Textbook and Additional References
- Summary

TEACHING TEAM INFO

Convenors Information:

- Name: Dr. Alejandro Guerra Manzanares
- E-mail: Alejandro.Guerra@nottingham.edu.cn
- Room: PMB435
- Office Hours: **TBA**
- Name: Dr. Wooi Ping Cheah
- E-mail: Wooi-Ping.Cheah@nottingham.edu.cn
- Room: PMB323
- Office Hours: **Wed 1pm-3pm**

Technician Information:

- Name: Ms. Jane Zhao
- E-mail: Jane.Zhao@nottingham.edu.cn

MODULE INFORMATION

- Class Sessions
 - Lectures and labs, ... and maybe some other stuff, too
 - We'll frontload a bit
 - Wednesdays, 11am-1pm, IAMET-326 (Weeks 1-3)
 - Thursdays, 4pm-6pm, DB-C05
 - Fridays, 11am-1pm, IAMET-406
- Labs (and Coursework)
 - ~5 main labs
 - ... more details soon!
- All materials on Moodle module page

COMP3052 Tentative Module Schedule 2025-2026 Spring Semester

Week	Week Commencing	Wednesday (11am-1pm) (IAMET-326)	Thursday (4pm-6pm) (DB-C05)	Friday (11am-1pm) (IAMET-406)
01 (24)	23-Feb	Introductions	Motivating Examples	Foundations
02 (25)	02-Mar	Authentications	Access Control	Firewalls
03 (26)	09-Mar	Network Security	Internet Security	<i>Lab 1: Intro to Kali</i>
04 (27)	16-Mar		Unix/Linux Security	<i>Lab 2: Passwords</i>
05 (28)	23-Mar		Windows Security	<i>Lab 3: Firewalls</i>
06 (29)	30-Mar		Intrusion Detection	Software Vulnerabilities
07 (30)	06-Apr		Data Security	<i>Lab 4: Packet Sniffing</i>
08 (31)	13-Apr		Crypto I	<i>Lab 5: Attack & Defend</i>
09 (32)	20-Apr		Crypto II & III	<i>Lab 6: Lab Assignment</i>
10 (33)	27-Apr		Crypto IV & V	Public Holiday
11 (34)	04-May	Metamorphic Security (2pm-4pm) (IAMET-406)	Revision / Q&A	
12 (35)	11-May			

ASSESSMENT

- 1 hour written examination 60 %
- Coursework 40 %
 - More details later ... but it will almost certainly be based on your experiences and reflections on the series of lab activities

ACTIVITY ...

- Come up with definitions for “security,” and “computer security,” and “security engineer”
- Why are we, as humans, concerned with these issues?

MOTIVATION

- People have protected their property and privacy for generations (Locks, Fences, Signatures, Seals, etc...)
- Big change
- Everything becoming electronic
- And security?
- What about the future?



ACTIVITY ...

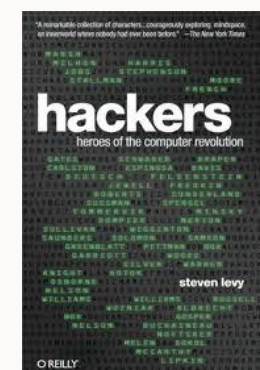
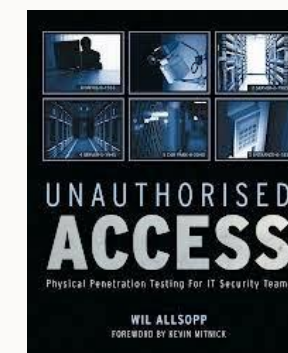
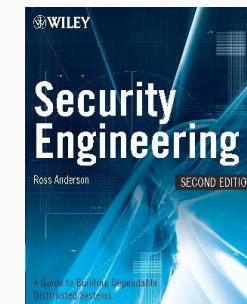
- List some points about what you *expect* to learn from this module, and what you *hope* to learn.
- Why?

LEARNING OUTCOMES

- What is computer/information security ?
- Why is it so important ?
- How can we evaluate and measure it ?
- How can we enforce it ?
- How can we minimise its risks ?
- The bad guy's point of view
- The victim's point of view

RESOURCES

- Core text:
 - Security Engineering – Ross Anderson 2nd/3rd edition (some available online, for free)
 - Computer Security – Dieter Gollmann 3d edition (Amazon)
- Additional Reading:
 - Secrets & Lies – Bruce Schneier
 - Unauthorised Access - Will Allsopp
 - Hackers - Steven Levy
- Module materials on Moodle



INTRODUCTION TO SECURITY



OUTLINE

- On Security
- Attacks and Attackers
- Security Management
 - Security Policies
 - Measuring Security
 - Standards
- Risk and Threat Analysis
 - Assets
 - Vulnerabilities
 - Threats
 - Risks
 - Countermeasures

SECURE SYSTEMS

- A secure system is one which does not exist...

An almost secure system is one which is locked up in a nuclear bunker within an air locked titanium safe and disconnected from anything else in the world.....and even such a system is not 100% secure!

- It is not about achieving complete security
- It is about minimising risk to systems
- Both from a technical, and social, point of view

WIKILEAKS SERVER BUNKER



- http://www.youtube.com/watch?feature=player_embedded&v=wn8pz1HLYp8

ON SECURITY

- Original focus on systems with **single, or few users**
- Today focus on **ubiquitous end systems**
- Systems interconnected by **networks**
- Danger of possible attacks from '**un-trustworthy**' nodes
- Both **remotely** as well as **locally** (insiders)
- Primarily a **management** issue!

ACTIVITY ...

- Based on your own impressions or knowledge, who are the “*attackers*”?
- What are the “*attacks*”?

ATTACKS AND ATTACKERS

- Landscape is changing
- Hackers -> Organised crime
- Website defacement -> Personal data harvesting
- Peer appreciation -> Earning money
- Viruses -> Trojans and Denial-of-Service attacks
- Complexity of our systems is increasing
- Our understanding of the system's intricacies can't keep up

SECURITY

- Reliability – Accidental failures
- Usability – Operating mistakes
- Security – Intentional failures

1. 'Security is a people problem'
2. Legal system defines boundaries of acceptable behaviour
3. Management responsible for security

SECURITY MANAGEMENT

- Management **responsible** for assets
- Security measures must have clear full **support** of senior management
- Security **awareness** programs
- **User** is not (usually) the enemy!
- **Developers** need even more awareness!

SECURITY POLICIES

- State what should be protected
- And how this should be achieved
- Security Policy Objective
- Organisational Security Policy
- Automated Security Policy

MEASURING SECURITY

- Very difficult
- Measures only exist for some aspects of security
- Product Security
- System Security
- Cost of an Attack
- Cost of Assets



RISK AND THREAT ANALYSIS

- Risk Analysis
 - All information assets
 - IT infrastructure
 - Perform during development



- Risk – **Possibility** of an incident or attack to cause **damage** to your enterprise
- Risk = **Assets** * **Vulnerabilities** * **Threat**

ACTIVITY ...

- What are assets, vulnerabilities, and threats?
 - Come up with definitions, and list some examples

ASSETS

- Software
 - Hardware
 - Data and Information
 - Reputation
-
- Identification easy, valuation difficult
 - Data, Information, Reputation – difficult to measure

VULNERABILITIES

- **Weaknesses** of a system that could be accidentally or intentionally **exploited** to damage assets
- Badly **configured** accounts
- Programs with **known flaws**
- Weak **access** control
- Weak **firewall** configuration
- Can be **rated** according to impact

THREATS

- Actions by adversaries who try to exploit vulnerabilities to damage assets
- Categorisation by damage done to assets
- Identification of source of attacks
- Analysis of attack execution (Attack Graphs)
- Can be rated according to likelihood
- Attack Graphs
 - formalised and structured
 - assessable, reproducible

RISK

- Quantitative Risk Analysis
 - + probability theory based on **mathematical** theory
 - - quality of results depends on **quality of inputs**
 - - not always **feasible**
- Qualitative Risk Analysis
 - + more **applicable**
 - - scaling based on **judgements** of security experts

COUNTERMEASURES

- Risk analysis **generates** recommended countermeasures
- **Up to date/continuous** risk analysis not always possible
- Baseline protection – security requirements for **typical cases** with recommended countermeasures

SUMMARY

- Current security landscape
- Management is vital to security
- How security can be measured
- What is Risk and how it is analysed

Read Anderson: Chapter 1